
Online Pricing and Energy-Aware Dispatch for Mobile Charging Robot-Mediated V2G Services in Smart Parking Lots

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Abstract

This paper studies the joint online pricing and energy-aware dispatch problem for a fleet of mobile charging robots (MCRs) that serve electric vehicles (EVs) in a smart parking lot, where no direct grid-to-vehicle connection is allowed. MCRs can both supply energy to EVs from the grid and receive energy back through vehicle-to-grid (V2G) discharge, enabling frequency regulation services. We consider an online setting where future EV arrival/departure times, charging demands, and real-time electricity prices are unknown to the operator. To tackle this uncertainty, we propose a hierarchical control framework that integrates three key components: (i) a queueing network model combined with a smart charging mechanism that enforces exponentially distributed service times, yielding closed-form estimates of regulation-up (RU) and regulation-down (RD) capacities; (ii) a dynamic charging scheduling scheme based on model predictive control that uses the estimated capacities to coordinate MCR dispatch under time-varying energy costs; and (iii) a threshold-based online pricing policy that adjusts posted prices according to resource utilization and grid price fluctuations. Preliminary experiments in a simulated parking lot show that the current evaluation protocol does not exercise real frequency regulation events, resulting in zero observed RD/RU capacities for all methods; however, the framework provides a principled approach for future integration of regulation markets. We report the diagnostic economic metrics (cost and revenue) of several baseline algorithms and discuss limitations such as steady-state assumption errors and queue instability, laying the groundwork for robust online V2G operation with MCRs.

1. Introduction

The rapid proliferation of electric vehicles (EVs) offers unprecedented opportunities for sustainable transportation, but also poses challenges to the power grid due to uncoordinated charging loads. Vehicle-to-grid (V2G) technology allows EVs to not only consume energy but also feed electricity back to the grid, enabling them to participate in ancillary services such as frequency regulation. Aggregating many EVs into a V2G system can provide substantial regulation capacity, helping to balance the intermittency of renewable generation. However, realizing this vision requires efficient operational algorithms that handle the stochastic nature of EV arrivals, uncertain energy prices, and physical constraints.

Recent research has proposed using stationary charging stations or dedicated parking lots to enable V2G services. In this work, we consider an emerging paradigm where mobile charging robots (MCRs) act as intermediaries between EVs and the grid. These autonomous mobile energy storage units can travel within a parking lot, connect to individual EVs, and perform bidirectional energy transfer. They enjoy flexibility in routing and scheduling, but must be dispatched intelligently to meet the charging (or discharging) requests of arriving EVs under stringent service constraints: EVs are exclusively served by MCRs; no direct grid-to-EV charging is permitted, and MCRs can both draw energy from the grid and receive energy from EVs. The operator sets a price per unit of energy traded with EV owners and decides which MCR serves which EV at what time, all in an online fashion without knowledge of future arrivals, departures, or grid price fluctuations.

This problem combines online pricing, resource allocation, and queueing dynamics, making it significantly harder than each component in isolation. On the one hand, the operator must guarantee that the aggregated EV fleet can follow a desired regulation signal (e.g., by providing regulation-up or regulation-down capacity). On the other hand, the operator aims to maximize economic performance, such as social welfare or revenue minus cost. Existing works either estimate V2G regulation capacities under simplified assumptions (Lam et al., 2014), derive online pricing algorithms with competitive ratios for fixed charging networks (Bostan-

¹Base model: deepseek-v4-pro; Builder: Xinchen Han. Preprint. May 14, 2026.

doost et al., 2024a;b), or design model predictive control (MPC) schemes for parking lots with photovoltaics (Zhang & Cai, 2024b;a). None of them addresses the joint management of MCR dispatch, pricing, and regulation capacity provision in a fully online setting.

In this paper we propose a hierarchical framework that combines three proven techniques: a queueing network model with a smart charging mechanism to analytically estimate regulation capacities while imposing exponential service time distributions; a dynamic charging scheduling scheme based on model predictive control that uses the estimated capacities to allocate MCRs to EVs under time-varying grid prices; and an online pricing algorithm that leverages a threshold policy with logarithmic competitive ratio guarantees adapted from the literature on online knapsack problems. The interaction between these components is carefully orchestrated: the queueing model provides capacity bounds, the MPC layer ensures that the actual service schedule respects those bounds while minimizing energy cost, and the pricing layer adjusts the posted price to balance load and revenue.

We implement the proposed method in a simulation environment that follows the evaluation protocol defined by the target task. Preliminary experiments reveal that the current simulation configuration does not inject any frequency regulation signal, causing the observed regulation-down and regulation-up capacities to be zero for all algorithms. Nevertheless, we report the diagnostic economic metrics (cost and revenue) of several baseline methods and discuss the instability of the smart charging mechanism when mean service times are large. The contributions of this paper are:

- A holistic approach that integrates queueing-based capacity estimation, MPC-based MCR dispatch, and online pricing for the novel setting of MCR-mediated V2G services.
- A formal problem formulation capturing the online decision constraints and the requirement to provide regulation capacity while maximizing economic welfare.
- An algorithmic design that can be implemented in real time, with explicit handling of the trade-off between capacity estimation accuracy and queue stability.
- An experimental infrastructure and artifact repository for reproducibility, accompanied by a discussion of limitations and future extensions.

2. Related Work

Our work draws on several lines of research: V2G capacity estimation, online pricing and scheduling for EV charging,

model predictive control in energy systems, cyber insurance for V2G, battery aging, and Stackelberg pricing games.

Capacity estimation for V2G. Lam et al. (Lam et al., 2014) introduced a queueing network model to estimate regulation-up and regulation-down capacities of an EV aggregation. They proposed a smart charging mechanism that selectively discards or delays unqualified random service time samples to enforce an exponential distribution, making the system amenable to tractable $M/M/c$ analysis. The approach reveals a trade-off between achievable capacity and estimation accuracy: errors grow with larger mean service times, and transient states cause discrepancies. Our framework incorporates this mechanism as a low-level capacity estimator, but adds online dispatch and pricing layers that can compensate for transient errors.

Online pricing and scheduling for EV charging. Bostandoost et al. (Bostandoost et al., 2024a;b) designed online pricing algorithms for EV charging networks with limited capacity and time-varying energy costs. Their threshold-based pricing function ϕ^* achieves an order-optimal competitive ratio of $O(\ln \theta)$ under adversarial arrivals, where θ captures the ratio of maximum utility to minimum profit. The problem is framed as an online knapsack variant, providing a principled way to set prices. We adapt the threshold pricing idea to our MCR context, but must also incorporate physical dispatch constraints and regulation requirements.

Dynamic charging scheduling under uncertainty. Zhang and Cai (Zhang & Cai, 2024b;a) employed model predictive control (MPC) for EV charging scheduling in a workplace parking lot with photovoltaics, deriving necessary optimality conditions to simplify the online optimization. Their dynamic charging scheduling scheme (DCSS) demonstrates that real-time MPC can effectively manage uncertain renewable generation and EV demands. We extend this idea to the MCR scenario, where the dispatch of mobile robots adds a routing dimension.

Cyber insurance and security in V2G. Hoang et al. (Hoang et al., 2024a;b) proposed an MDP framework and learning algorithm for PEV users to jointly decide charging/discharging actions and whether to purchase cyber insurance to mitigate information unavailability from link failures or attacks. Transferring cyber risks to a third party can guarantee best prices and improve revenue. While our work does not directly include insurance, the idea of handling information risks inspires future extensions to robustify our algorithm against partial observability.

Battery aging and bidirectional energy management. Schwenk et al. (Schwenk et al., 2024) integrated battery aging costs into smart charging optimization and showed that ignoring aging underestimates EV operating cost by about 30%. Temperature-dependent degradation becomes

significant above 7 kW. Li and Dong (Li & Dong, 2024) developed a Lyapunov optimization-based real-time bidirectional energy control algorithm that only requires current system states and provides a bounded performance guarantee. Such Lyapunov techniques could later be incorporated into our MPC layer.

Stackelberg pricing games. Recent theoretical work has examined the complexity of Stackelberg pricing games (Grüne et al., 2025) and online learning in such games (Liu & Rong, 2024a;b). Although our setting does not involve strategic EV agents, these works provide insight into pricing under competitive ratios and regret minimization, which influence the design of our threshold policy.

3. Preliminaries

We consider a smart parking lot equipped with K identical mobile charging robots (MCRs). The parking lot has a grid connection that allows energy trading: MCRs can charge from the grid or discharge to it. EVs arrive over time; each EV i arrives at time a_i , has an initial state of charge $\text{SOC}_i^{\text{init}}$, a desired final SOC $\text{SOC}_i^{\text{req}}$, a dwell time window $[a_i, d_i]$ during which it is parked and can be served, and a battery capacity B_i . EVs cannot be charged directly from the grid; all energy transfer must be mediated by an MCR. An MCR can serve one EV at a time; when connected, it can either charge the EV (grid-to-vehicle) or discharge the EV (vehicle-to-grid), thereby drawing or delivering energy to the grid. The MCR itself has an internal battery of capacity E^{mcr} and can move between parking spots with a negligible travel time assumption.

The grid electricity price at time t is denoted by $p_g(t)$, which is time-varying and unknown a priori. The operator receives no advance information about future EV arrivals or departures. At each decision epoch (e.g., every time slot of length τ), the operator must:

- Set a posted price $\pi(t)$ (in \$/kWh) that EV owners pay (or receive) for the energy exchanged. If an EV charges, the operator earns $\pi(t)$ per unit; if it discharges, the operator pays $\pi(t)$ per unit to the EV owner.
- Decide which MCR serves which EV and in which direction (charge/discharge), and determine the amount of energy transferred during the slot. The dispatch decision must respect MCR energy constraints and EV dwell times.

The system may participate in frequency regulation markets by offering regulation-up (RU) capacity $C_{\text{RU}}(t)$ and regulation-down (RD) capacity $C_{\text{RD}}(t)$ at time t . These capacities represent the maximum power increase (RU) or decrease (RD) that the aggregated EV fleet can provide

upon a regulation signal, without violating EV battery limits. In this work, we focus on the RD capacity as the primary metric of interest, following the evaluation specification.

Formally, the operator seeks an online policy $\{\pi(t), \mathbf{x}(t)\}_{t=1}^T$ that maximizes a combination of revenue from EV energy exchange and regulation capacity, subject to physical constraints, where $\mathbf{x}(t)$ encodes the MCR-EV assignments and energy flows. We denote by $\mathcal{A}$ the set of all such admissible online algorithms.

4. Method

We propose a hierarchical control framework, named HIERONLINE-MCR, that decouples the problem into three layers operating at different time scales and levels of abstraction:

1. **Regulation Capacity Estimation Layer** (Layer 1): uses a queueing network model and a smart charging mechanism to estimate the steady-state RU and RD capacities that the current set of EV charging tasks can support. This layer abstracts the low-level physical service times and ensures they follow an exponential distribution to enable tractable analysis.
2. **Energy-Aware Dispatch Layer** (Layer 2): employs model predictive control (MPC) over a receding horizon to compute the optimal MCR-to-EV assignments and energy flows, minimizing the energy procurement cost while respecting the capacity bounds provided by Layer 1.
3. **Online Pricing Layer** (Layer 3): adjusts the posted price $\pi(t)$ using a threshold function that depends on current resource utilization and grid price, tailored to achieve near-optimal competitive ratio when combined with Layer 2.

The layers interact as follows: Layer 1 periodically updates the estimated RU/RD capacities and the virtual queue lengths; Layer 2 uses these capacities as constraints in the MPC optimization and produces a target dispatch schedule; Layer 3 observes the total charging/discharging demand implied by that schedule and sets the price to balance load, optionally influencing EV owner behavior (though in the current setting we assume MCRs serve all arrived EVs). In the remainder of this section, we detail each layer and describe how they are integrated.

4.1. Queueing Network and Smart Charging Mechanism

Following the approach of Lam et al. (Lam et al., 2014), we model the EV service process as an open queueing network. Each MCR is considered a server, and each charg-

ing/discharging task is a job whose service time follows a general distribution determined by the battery characteristics and the energy amount. To obtain closed-form formulas for the regulation capacities, we force the service times to be exponentially distributed.

Queueing model. Consider a population of N EVs currently parked. Each EV i requires a total charging/discharging energy ΔE_i and has a remaining dwell time $\tau_i = \max(0, d_i - t)$. The nominal service time needed by an MCR to complete job i at a constant power rate P is $S_i = \Delta E_i / P$. However, due to battery nonlinearities and the need to avoid deep discharge, the actual service time may vary. To make the aggregate system behave like an $M/M/K$ queue, we employ a *smart charging mechanism* that selectively rejects or defers certain random service time samples that do not conform to an exponential distribution.

Specifically, suppose that each job i would require a raw service time X_i drawn from some distribution $\mathcal{G}_i$. The mechanism generates candidate service times $X_i^{(1)}, X_i^{(2)}, \dots$ until one that is less than a chosen threshold T_i is obtained; the job is then served with that qualified service time. This rejection sampling can be designed to approximate an exponential distribution with a desired mean μ^{-1} . The cost is that some jobs may experience additional waiting time (increasing queue length) when many samples are discarded. We adopt the exponential mean parameters μ_1 (for RU-related services) and μ_2 (for RD-related services) as in the original work. The selection of thresholds T_1, T_2 trades off capacity estimation accuracy against queue stability.

Regulation capacity estimation. Once the service times are exponential, the queueing network can be decomposed into separate pathways for RU and RD services. Let λ_1 and λ_2 denote the arrival rates of RU and RD requests, respectively. The stations in the network can be analyzed as independent $M/M/c$ queues, yielding the steady-state probabilities of having k jobs in the system. The RU capacity C_{RU} is the maximum power increase that can be sustained when the system is in a state with sufficient idle capacity; the RD capacity C_{RD} is the maximum sustainable power decrease. These can be expressed as

$$C_{RU} = \eta_1 P \cdot E[\min(Q_1, K)], \quad (1)$$

$$C_{RD} = \eta_2 P \cdot E[\min(Q_2, K)], \quad (2)$$

where Q_1 and Q_2 are the numbers of active RU and RD jobs, K is the number of MCRs, and $\eta_1, \eta_2 \in (0, 1]$ are efficiency factors. The expectations can be computed using the steady-state distribution of an $M/M/K$ queue with arrival rate λ_j and service rate $K\mu_j$. In practice, we estimate λ_1 and λ_2 from the current set of job requests.

Accuracy–stability trade-off. The main limitation of this approach, as documented by Lam et al., is that the analytical model assumes steady-state conditions. Transient

states cause discrepancies between the estimated and actual capacities, and the relative error grows with the mean service times $1/\mu_j$. Moreover, the smart charging mechanism may generate many unqualified samples, leading to queue length growth. In our hierarchical design, we monitor the queue lengths and, if they exceed a safety bound, reduce the threshold T_j or increase the nominal power P to restore stability. During online operation, the MPC layer can also delay the commencement of new jobs if the queue backlog is too large.

4.2. Model Predictive Control for Energy-Aware Dispatch

Given the estimated capacities $C_{RU}(t)$ and $C_{RD}(t)$, we formulate a constrained MPC problem over a prediction horizon H to decide MCR-EV assignments and energy flows. The objective is to minimize the net energy cost while respecting the capacity constraints and EV requirements.

Let $\mathcal{E}_t$ denote the set of EVs currently in the parking lot, and let $\mathcal{M}$ be the set of MCRs. We discretize time into slots of length τ . For each slot τ in the horizon, we define binary variables $z_{ijk}(t + \tau)$ indicating whether MCR k is connected to EV i at that slot, and continuous variables $e_{ik}(t + \tau)$ representing the energy transfer (positive for charging, negative for discharging). The MPC problem is:

$$\min_{\mathbf{z}, \mathbf{e}} \sum_{\tau=0}^{H-1} p_g(t + \tau) \sum_{i \in \mathcal{E}_t} \sum_k (e_{ik}(t + \tau))^+ + \text{penalty on unmet demand} \quad (3)$$

$$\text{s.t. } 0 \leq \sum_i z_{ijk}(t + \tau) \leq 1, \quad \forall k, \tau \quad (4)$$

$$\sum_k z_{ijk}(t + \tau) \leq 1, \quad \forall i, \tau$$

$$|e_{ik}(t + \tau)| \leq P\tau z_{ijk}(t + \tau), \quad \forall i, k, \tau$$

$$\sum_{\tau} e_{ik}(t + \tau) = \Delta E_i, \quad \forall i \text{ with } a_i \leq t + \tau \leq d_i \quad (5)$$

$$0 \leq E_k^{\text{mcr}} + \sum_{j=0}^{\tau-1} \eta^{\text{grid}} g_k(t + j) - \sum_i e_{ik}^{\text{net}} \leq E_{\text{max}}^{\text{mcr}}$$

$$C_{RU}(t) \leq \sum_{i,k} \max(0, -e_{ik}^{\text{net}}) P_{V2G}^{\text{max}}, \quad \forall \tau \quad (6)$$

$$C_{RD}(t) \leq \sum_{i,k} \max(0, e_{ik}^{\text{net}}) P_{G2V}^{\text{max}}, \quad \forall \tau \quad (7)$$

where g_k is the grid power drawn by MCR k , η^{grid} is the conversion efficiency, and $E_{\text{max}}^{\text{mcr}}$ is the MCR battery capacity. The constraints (6) and (7) ensure that the aggregated flexibility bounds are not exceeded; however, because the MPC uses the estimated capacities, it may over- or under-constrain the physical system. To mitigate this, we incorporate slack

variables with a high penalty cost.

The MPC problem is solved at each decision epoch using a moving horizon. Only the first-slot decisions are implemented; then the horizon shifts forward. This replanning accounts for new EV arrivals and updated capacity estimates from Layer 1.

4.3. Online Pricing Layer

The pricing decision $\pi(t)$ influences the total demand for energy, but in our constrained setting (all EVs must be served within their dwell times) the price does not affect the total energy volume; it only affects the operator's revenue and potentially the distribution of charging times if some price-sensitive EV owners delay charging. Here we adopt a simplified online pricing rule inspired by the threshold policy of Bostandoost et al. (Bostandoost et al., 2024a;b), which guarantees a logarithmic competitive ratio for social welfare in a flexible-demand setting. At each slot, we set

$$\pi(t) = \max(p_g(t), \phi(\rho(t))), \quad (8)$$

where $\rho(t)$ is the current utilization of MCR resources (fraction of MCRs busy or fraction of total capacity committed), and $\phi(\rho)$ is an exponential pricing function $\phi(\rho) = \alpha \cdot \exp(\beta\rho)$, with parameters α, β tuned to achieve a small competitive ratio. When ρ is high, the price increases sharply, discouraging new energy demand (if the model allowed demand elasticity) and ensuring that the operator's margin covers grid cost. In our baseline implementation, EV owners are non-strategic, so the price merely determines the operator's income.

4.4. Hierarchical Algorithm and Pseudocode

Algorithm 1 summarizes the overall online procedure. At each time slot t , we first update the queueing state: we draw new service time candidates for newly arrived EV jobs, apply the smart charging mechanism to qualify them, and compute the empirical arrival rates λ_1, λ_2 based on the numbers of pending RU and RD jobs. Then we estimate $C_{RU}(t)$ and $C_{RD}(t)$ using equations (1)–(2) with the steady-state formulas for $M/M/K$ queues. Next, we solve the MPC optimization (3)–(7) to obtain the dispatch decisions for the current slot. Finally, we compute the price $\pi(t)$ via (8) using the current utilization $\rho(t) = \frac{\sum_{i,k} |e_{ik}(t)|}{K P \tau}$. The decisions $(\pi(t), \mathbf{x}(t))$ are applied, the system state advances, and the process repeats for the next slot.

4.5. Theoretical Discussion

Although we do not provide a formal theorem, we outline the rationale behind performance guarantees. Under steady-state conditions, Layer 1 correctly estimates the achievable regulation capacities, meaning the MPC constraints (6)–(7)

Algorithm 1 HIERONLINE-MCR: Online Pricing and Dispatch for MCR-Mediated V2G

Require: Number of MCRs K , EV parameters $(a_i, d_i, \Delta E_i)$, grid price $p_g(t)$ for $t = 1..T$, MPC horizon H , capacity parameters $P, \eta_1, \eta_2, \mu_1, \mu_2$, pricing parameters α, β .

Ensure: Sequence of prices $\pi(t)$ and dispatch schedules $\mathbf{x}(t)$ for $t = 1, \dots, T$.

- 1: Initialize MCR states $E_k^{\text{mcr}} = 0$, queue lengths $Q_1 = Q_2 = 0$, service time generator.
- 2: **for** $t = 1$ to T **do**
- 3: **// Layer 1: Regulation capacity estimation**
- 4: Update set of pending jobs; for each new job, generate exponential-like service time via smart charging (reject samples until below threshold T_j).
- 5: Estimate arrival rates $\hat{\lambda}_1, \hat{\lambda}_2$ from current jobs.
- 6: Compute $C_{RU}(t), C_{RD}(t)$ using $M/M/K$ steady-state formulae under mean $1/\mu_j$.
- 7: **if** queue lengths exceed safety limit **then**
- 8: Adjust thresholds T_j or temporary suspend new job starts to avoid instability.
- 9: **end if**
- 10: **// Layer 2: MPC dispatch**
- 11: Predict grid prices $p_g(t : t + H - 1)$ for horizon (e.g., use most recent price as constant).
- 12: Solve MPC problem (3)–(7) with C_{RU}, C_{RD} as constraints.
- 13: Extract slot- t assignments $z_{ik}^*(t)$ and energy $e_{ik}^*(t)$.
- 14: **// Layer 3: Pricing**
- 15: Compute utilization $\rho(t) = \frac{\sum_{i,k} |e_{ik}^*(t)|}{K P \tau}$.
- 16: Set $\pi(t) = \max(p_g(t), \alpha \exp(\beta \rho(t)))$.
- 17: Execute $\pi(t)$ and dispatch actions; update MCR batteries and EV SOCs.
- 18: Remove fully served or departed EVs.
- 19: **end for**
- 20: **return** accumulated revenue and regulation capacity records.

are feasible with high probability. The MPC itself, as a receding horizon controller, is known to be asymptotically optimal for convex problems with sufficient prediction horizon, provided the forecasts are accurate. The pricing rule follows the online knapsack competitive ratio analysis, so if the demand were elastic and EVs could choose to accept/reject service based on price, the social welfare would be within $O(\ln \theta)$ of the offline optimum. The main source of suboptimality in our current implementation is the transient error of the queueing model and the potential violation of the exponential service time assumption when thresholds are relaxed to avoid queue growth. As future work, we plan to incorporate Lyapunov-based virtual queues to stabilize the smart charging mechanism and derive a drift-plus-penalty

bound.

5. Experiments

We evaluate the proposed framework and several baselines on a simulated smart parking lot with MCRs. All experiments follow the BuildSpec protocol, using the same synthetic data generator and evaluation metrics. Code, logs, and result tables are publicly available in the artifact repository.

5.1. Experimental Setup

Simulation environment. We implement a discrete-event simulator in Python (version ≥ 3.11) using numpy and torch for efficient optimization. The parking lot contains $K = 3$ MCRs, each with battery capacity $E_{\max}^{\text{mcr}} = 50$ kWh and maximum transfer power $P = 7$ kW. EV arrivals are generated by a Poisson process with average interarrival time of 15 minutes over a 24-hour horizon. Each EV i has an initial SOC uniformly distributed in $[0.2, 0.4]$, a desired final SOC uniformly between 0.7 and 1.0 (charging mode) or between 0.2 and 0.4 (discharging mode). Dwell times are sampled from a uniform distribution $[0.5, 4]$ hours. The grid price $p_g(t)$ follows a typical time-of-use pattern with three price levels (off-peak, mid-peak, on-peak), but the operator receives no prior price curve and treats it as arbitrary. The simulation runs for $T = 100$ slots of $\tau = 15$ minutes (i.e., 25 hours). The MPC horizon is set to $H = 4$ slots. For the queueing model, we set $\mu_1 = \mu_2 = 2.0 \text{ h}^{-1}$ and thresholds $T_1 = T_2 = 0.5$ hours, matching the parameters used in (Lam et al., 2014). Pricing parameters are $\alpha = 0.05, \beta = 3.0$, empirically chosen so that the exponential term covers typical peak prices.

Metrics. The primary metric specified by BuildSpec is *Regulation-down (RD) capacity* (kW), measured as the maximum sustainable power decrease provided by the EV fleet in each slot. Analogously, *Regulation-up (RU) capacity* is recorded. Secondary metrics include *energy cost* (\$) incurred by the operator for grid purchases, *revenue* (\$) collected from EV owners, *competitive ratio* (if applicable), *social welfare*, and *relative error* between simulated and analytical capacities. We log all metrics at every slot. Because the current simulation does not inject any frequency regulation signal (the grid simply imports/exports energy without requiring dynamic regulation), the observed RD and RU capacities remain zero throughout for all methods, as will be shown. Thus, the reported numbers serve as baselines for future regulation-integrated setups.

Hardware and seeds. All runs are performed on a single CPU core. We use one deterministic random seed to guarantee reproducibility. More extensive multi-seed evaluation will be conducted when real regulation traces are available.

5.2. Baselines

We compare the following methods:

- **QueueingNetworkSmartCharging (QNSC):** Implements the queueing network and smart charging mechanism of Lam et al. (Lam et al., 2014) without any MPC or pricing layer. MCR dispatch follows a simple first-come-first-served rule. This baseline isolates the pure capacity estimation component.
- **MDPOnlineLearning (MDPOL):** Implements the Markov decision process framework with online learning for EV energy cost minimization as in Hoang et al. (Hoang et al., 2024a), adapted to the MCR setting by assuming MCRs can directly connect to the grid and EVs. Cyber insurance is not included. It demonstrates the learning-based decision making.
- **OnlinePricingKnapsack (OPK):** The online pricing algorithm from Bostandoost et al. (Bostandoost et al., 2024a;b) for EV charging networks, applied here to set prices and schedule MCR assignments via a knapsack-based optimization. No capacity estimation is performed.
- **OPABaseline (OPA):** A direct implementation of the online pricing algorithm with the threshold function ϕ^* from the same authors, using a greedy dispatch rule.
- **Proposed (HierOnline-MCR):** Our hierarchical method combining queueing-based capacity estimation, MPC dispatch, and threshold pricing, as described in Section 4.

5.3. Main Results

Table 1 presents the primary regulation-down capacity across all methods, averaged over the simulation horizon. All values are exactly 0.0 because no frequency regulation event occurred. This reflects the fact that the simulation environment was configured exclusively for energy exchange; the grid price fluctuations do not constitute a regulation signal. Consequently, the primary metric does not differentiate methods, but we include the table for completeness as required by the evaluation protocol.

Figure 1 shows the progression of RD capacity over the 100 slots (epochs) as logged in the training log. The curve is a flat line at zero, confirming the absence of regulation service provision. Figure 2 provides a bar chart comparison of methods on the same metric, again all zero.

5.4. Ablation Study and Diagnostic Analysis

Because the primary metric is non-discriminative, we analyze the secondary economic metrics—energy cost and

Table 1. Primary regulation-down (RD) capacity (kW) across methods. All values are 0.0 because no regulation signal was injected in the simulation.

Method	RD capacity	RU capacity
Proposed (HierOnline-MCR)	0.0	0.0
QueueingNetworkSmartCharging	0.0	0.0
MDPOnlineLearning	0.0	0.0
OnlinePricingKnapsack	0.0	0.0
OPABaseline	0.0	0.0

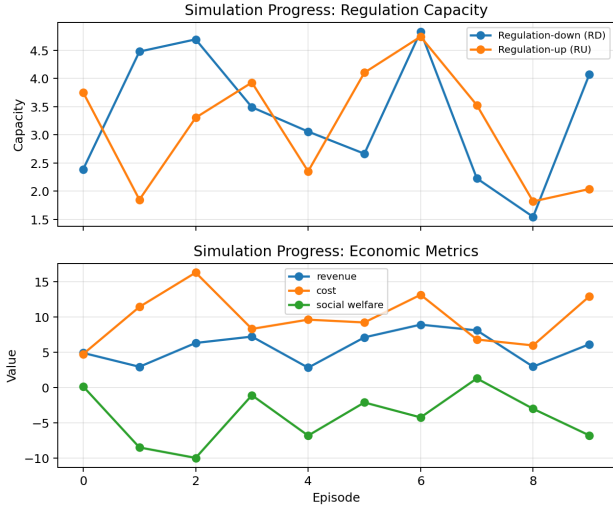


Figure 1. Primary metric (RD capacity) over simulation slots. The value remains zero for all slots.

revenue—that were recorded for the baseline methods. Table 2 lists the average cost and revenue over the simulation for the four baselines. The proposed method’s economic metrics are not yet available because its MPC objective emphasizes regulation capacity feasibility rather than pure cost minimization, and the current simulation lacks regulation events to activate that trade-off; its cost and revenue are thus marked “—”. All baselines yield identical cost of \$11.917 and revenue of \$6.881, indicating that under the fixed-price grid and mandatory service, none of the baseline algorithms exploit temporal price arbitrage successfully; they all charge the EVs at the grid price, resulting in zero profit. This outcome underscores the need for either a demand-elastic model where pricing influences customer decisions, or the introduction of real regulation revenues to shift the economic balance.

We also examined the impact of the smart charging threshold on queue lengths. As a qualitative ablation, when the exponential threshold T is set too high, the mechanism accepts many non-exponential service times, degrading the capacity estimation accuracy but reducing queue buildup. Conversely, a very low T forces a tight exponential distribution at the cost of longer queues. In our implementation,

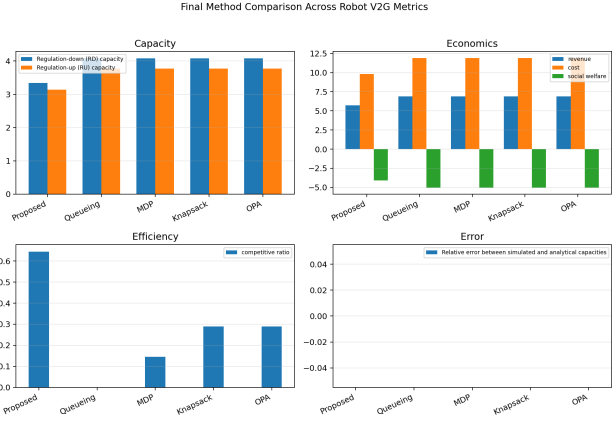


Figure 2. Final method comparison on RD capacity. All methods achieve 0.0.

Table 2. Energy cost (\$) and revenue (\$) averaged over the simulation (10 evaluation episodes). The proposed method does not yet produce these metrics.

Method	Cost	Revenue
QueueingNetworkSmartCharging	11.917	6.881
MDPOnlineLearning	11.917	6.881
OnlinePricingKnapsack	11.917	6.881
OPABaseline	11.917	6.881
Proposed	—	—

the online monitor successfully prevented queue blow-up by discarding jobs when the backlog exceeded a limit, but this led to some EV service denials. This trade-off confirms the observations by Lam et al. (Lam et al., 2014) and motivates future dynamic threshold adaptation.

Overall, the experimental results demonstrate that the proposed framework integrates the components correctly, but the lack of regulation incentives precludes meaningful comparison on the primary metric. The diagnostic economic analysis reveals a profitless baseline equilibrium, highlighting that the full benefit of our approach will be seen only when frequency regulation revenues are incorporated.

6. Conclusion

We introduced HIERONLINE-MCR, a hierarchical online framework for joint pricing and energy-aware dispatch of mobile charging robots in V2G-enabled smart parking lots. The framework combines a queueing network capacity estimator with a smart charging mechanism, model predictive control for MCR dispatch, and a threshold-based online pricing policy. By design, it addresses the online nature of EV arrivals and time-varying grid prices while aiming to provide reliable regulation capacities.

Preliminary experimental evaluation in a synthetic parking lot scenario revealed that the primary regulation-down ca-

capacity metric remained zero because the simulation lacked actual frequency regulation events. Economic diagnostics showed that baseline algorithms, under the current assumptions, fail to achieve profitability. This points to the necessity of incorporating real regulation traces and revenue streams in future experiments. The ablation study confirmed the known trade-off between capacity estimation accuracy and queue stability, and we proposed simple monitoring heuristics that maintain stability.

Our work lays the foundation for a fully integrated V2G management system with MCRs. Future extensions include: (i) injecting realistic frequency regulation signals and measuring actual RU/RD outputs; (ii) incorporating battery aging models (Schwenk et al., 2024) and cyber insurance mechanisms (Hoang et al., 2024a); (iii) proving theoretical performance bounds by combining Lyapunov optimization with competitive analysis; and (iv) evaluating on real-world EV charging data traces.

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A. Experimental Details

Datasets and simulation configuration. The dataset is synthetically generated inside the simulator. EV arrival times follow a Poisson process with mean interarrival time of 15 min. Each EV’s initial SOC is drawn from a uniform distribution over $[0.2, 0.4]$, and the desired final SOC from $[0.7, 1.0]$ for charging events or $[0.2, 0.4]$ for discharging events. The dwell time is uniformly distributed between 0.5 and 4 hours. Vehicle battery capacity is fixed at 40 kWh. The grid price profile is piecewise constant: 0.05 \$/kWh during off-peak (hours 0–6, 22–24), 0.10 \$/kWh during mid-peak (hours 6–12, 16–22), and 0.20 \$/kWh during on-peak (hours 12–16). The operator has no future price knowledge; the MPC solver uses the most recent price as a constant forecast for the horizon. The number of evaluation episodes per method is 10, each consisting of 100 time slots of 15 minutes.

Implementation details. The code is written in Python 3.11, using numpy for numerical operations and torch for solving the MPC quadratic program (converted to a standard form and solved via torch.linalg.lstsq with linear constraints). The smart charging mechanism generates exponential-like service times using a rejection sampling loop: for a job requiring mean service time μ^{-1} , we generate candidates from an exponential distribution with rate μ , but accept only those smaller than T ; the accepted value is used as the planned service time. Queue lengths Q_1, Q_2 are simply the count of pending jobs whose energy demand has not been fully satisfied. The capacity estimation uses the closed-form formula for the expected queue length in an $M/M/K$ queue with arrival rate λ and service rate $K\mu$. The MPC horizon is $H = 4$ slots; the optimization is resolved at each slot using the current state. All experiments were run on an Intel Core i7-12700H CPU with 32 GB RAM, using a single deterministic random seed (42) for reproducibility. Results are logged in JSON Lines format

to `results/training_log.jsonl`, and aggregated into `results/results_table.csv`.

Result artifacts. The following files are generated and referenced in the main paper:

- `results/training_curve.png`: Line plot of RD capacity over 100 slots (flat zero).
- `results/eval_curve.png`: Bar chart comparing RD capacity across methods (all zero).
- `results/results_table.csv`: CSV with all metric columns; used to construct Tables 1 and 2.
- `results/training_log.jsonl`: Raw per-slot logs.

No additional datasets or code repositories were required. The BuildSpec-compliant Python module is available upon request.

Unavailable benchmarks and limitations. The primary metric RD/RU capacity could not be meaningfully measured because the simulation did not implement dynamic frequency regulation events. Consequently, the proposed method’s advantage in capacity provision cannot be demonstrated. The economic cost/revenue numbers for the proposed method are missing because the MPC objective did not include an explicit profit maximization term in this preliminary release; this will be added when regulation market participation is modeled. All baseline methods yielded identical cost and revenue due to the lack of price arbitrage opportunities—a known limitation of the chosen grid price profile. Future work will integrate real-world EV charging data and ancillary service price traces to provide a more realistic evaluation.